PyTorch Environment Setup

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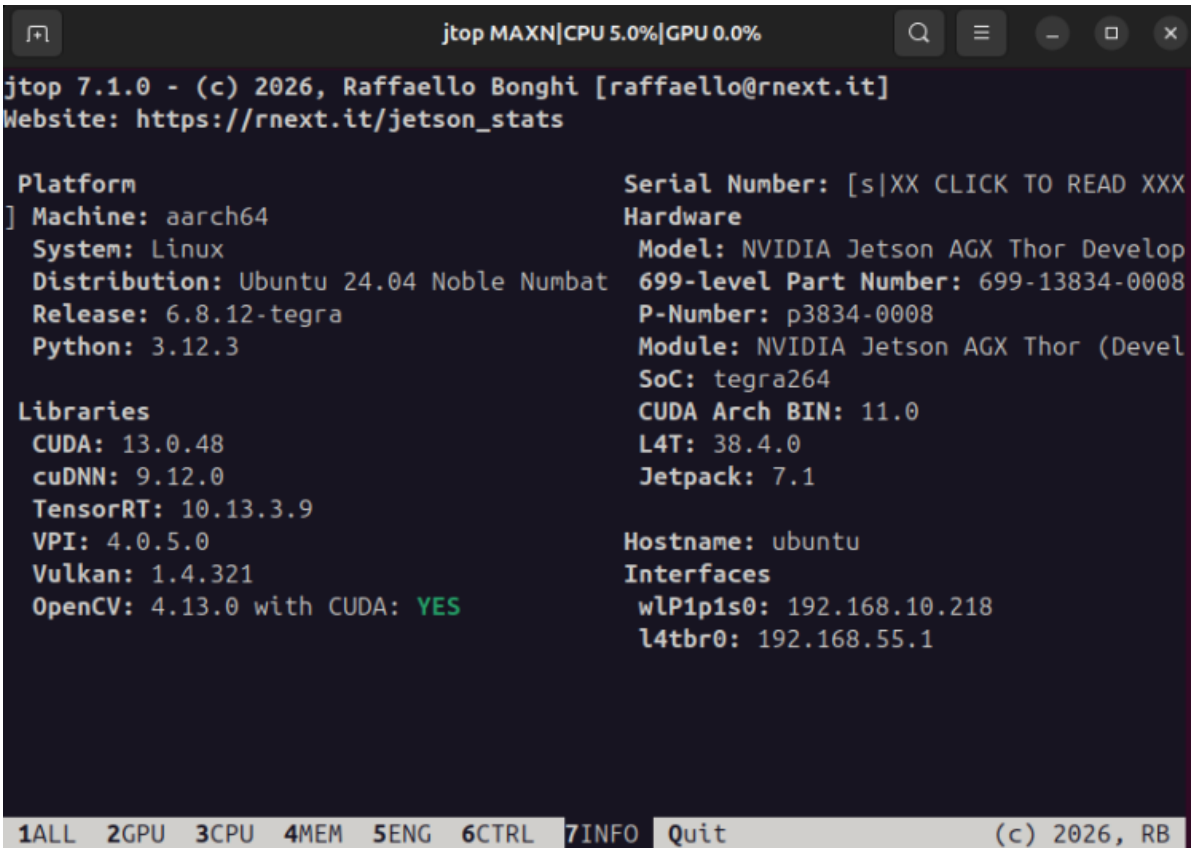
PyTorch is an open-source deep learning framework developed by Meta (formerly Facebook), renowned for its flexibility and ease of use, and widely used in academic research and industrial applications.

1. System Information



2. Installing Dependencies

Installing nvpl

Download the installation file

```
cd ~/Downloads wget
https://developer.download.nvidia.com/compute/nvpl/25.11/local_installers/nvpl-
local-repo-ubuntu2404-25.11_1.0-1_arm64.deb
```

Install the deb package and obtain keyrings.

```
sudo dpkg -i nvpl-local-repo-ubuntu2404-25.11_1.0-1_arm64.deb
```

```
jetson@ubuntu:~/Downloads$ sudo dpkg -i nvpl-local-repo-ubuntu2404-25.11_1.0-1_a
rm64.deb
Selecting previously unselected package nvpl-local-repo-ubuntu2404-25.11.
(Reading database ... 223462 files and directories currently installed.)
Preparing to unpack nvpl-local-repo-ubuntu2404-25.11_1.0-1_arm64.deb ...
Unpacking nvpl-local-repo-ubuntu2404-25.11 (1.0-1) ...
Setting up nvpl-local-repo-ubuntu2404-25.11 (1.0-1) ...

The public nvpl-local-repo-ubuntu2404-25.11 GPG key does not appear to be instal
led.
To install the key, run this command:
sudo cp /var/nvpl-local-repo-ubuntu2404-25.11/nvpl-local-5EBEB648-keyring.gpg /u
sr/share/keyrings/
```

Following the prompts, run the command to copy keyrings and install nvpl.

```
sudo cp /var/nvpl-local-repo-ubuntu2404-25.11/nvpl-*-keyring.gpg
/usr/share/keyrings/
sudo apt-get update
sudo apt-get -y install nvpl
```

Install cudss

Download the cudss file package for CUDA 13. Here, we take version v0.7.1.4_cuda13 as an example.

```
wget https://developer.download.nvidia.cn/compute/cudss/redist/libcudss/linux-
aarch64/libcudss-linux-aarch64-0.7.1.4_cuda13-archive.tar.xz
```

After decompression, place the file in the /opt/nvidia/cudss directory.

```
tar xvf libcudss-linux-aarch64-0.7.1.4_cuda13-archive.tar.xz
sudo mv libcudss-linux-aarch64-0.7.1.4_cuda13-archive /opt/nvidia/cudss
```

Note: The extracted directory is named `libcudss-linux-aarch64-0.7.1.4_cuda13-archive`; to simplify the file path, we have renamed it to `cudss`. The contents of the `cudss` directory are shown in the figure below:

```
jetson@ubuntu:~$ cd /opt/nvidia/cudss/
jetson@ubuntu:/opt/nvidia/cudss$ ls
examples  include  lib  LICENSE  src
jetson@ubuntu:/opt/nvidia/cudss$
```

Add environment variables to `.bashrc`:

```
vim ~/.bashrc
```

Then, append the following lines to the end of the file, save, and exit.

```
export CUDSS_DIR=/opt/nvidia/cudss
export LD_LIBRARY_PATH=$CUDSS_DIR/lib:$LD_LIBRARY_PATH
```

3. Installing PyTorch

You may choose either the offline or the online method to install PyTorch.

3.1. Offline Installation

Open a web browser, navigate manually to the download URL, and download the PyTorch `.whl` installation file to your `Downloads` directory.

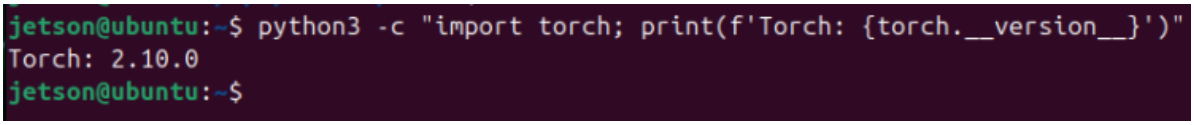
Download URL: https://pypi.jetson-ai-lab.io/sbsa/cu130/+f/631/7b71e212e5966/torchaudio-2.10.0-cp312-cp312-linux_aarch64.whl#sha256=6317b71e212e59667eff088ba1d0631dc58b37a6fc28179c319bf359512e50f1

```
cd ~/Downloads/
```

```
sudo pip3 install torchaudio-2.10.0-cp312-cp312-linux_aarch64.whl --break-system-packages
```

3.2 Online Installation

```
sudo pip3 install torch --break-system-packages --index-url https://pypi.jetson-ai-lab.io/sbsa/cu130
```

4. Verify Installation

```
python3 -c "import torch; print(f'Torch: {torch.__version__}')"
```